WEB DESIGN EXAM REVIEW

Web Colors: RGB and Hex Codes

How a computer describes a color with numbers



VIDEO 1 · RGB COLOR MODEL

Screens make color by mixing red, green, and blue light

Light mixed	Color you see
Red + Green	Yellow
Red + Blue	Magenta
Green + Blue	Cyan
Red + Green + Blue (all full)	White
No light at all	Black



This is called **additive** color: adding more light makes the color brighter. Paint and printers work the opposite way.

VIDEO 2 · RGB VALUES

Each color gets a number from 0 to 255

`rgb(red, green, blue)`

0 = none of that color

255 = full amount

256 levels per color

```
rgb(255, 0, 0) = red
rgb(0, 0, 0) = black
rgb(255, 255, 255) = white
```

256 × 256 × 256 = about **16.7 million** possible colors.

VIDEO 2 · HEX CODES

A hex code is the same three numbers, written in pairs

#FF8800

starts every hex code

6 digits: 2 for red, 2 for green, 2 for blue

Order is always **RR GG BB**

VIDEO 2 · COUNTING IN HEX

Hex counts to 15 with one digit, using letters

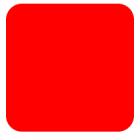
0 1 2 3 4 5 6 7 8 9 A B C D E F

Hex pair	Normal number	Meaning
00	0	None
80	128	About half
FF	255	Full

Hex is **base 16** (hexadecimal). A = 10, B = 11, C = 12, D = 13, E = 14, F = 15.

VIDEO 2 · READING A HEX CODE

Look for the biggest pair



`#FF0000 → all red`



`#0000FF → all blue`



`#808080 → equal = gray`

`#FFFF00 → red + green = yellow`

When all three pairs are equal, the color is a gray: low = dark, high = light.

VIDEO 3 · RECAP

One color, three ways to write it

Name	Hex	RGB
red	#FF0000	rgb(255, 0, 0)
lime	#00FF00	rgb(0, 255, 0)
blue	#0000FF	rgb(0, 0, 255)
white	#FFFFFF	rgb(255, 255, 255)
black	#000000	rgb(0, 0, 0)
gray	#808080	rgb(128, 128, 128)

Writing colors in CSS

```
h1 { color: navy; }
p { color: #333333; }
body { background-color: rgb(240, 248, 255); }
.box { background-color: rgba(0, 0, 255, 0.5); }
```

color = text color
background-color = the box behind it

rgba adds a 4th number for see-through: 0 = clear, 1 = solid

Short hex: #F00 is the same as #FF0000

Practice: what color is it?

1. `#00FF00`
2. `#FFFFFF`
3. `#FF0000`
4. `#FFFF00`
5. `rgb(0, 0, 0)`
6. `rgb(255, 0, 255)`
7. `#333333`
8. Write blue in hex

For the exam

If the question says…	Remember
Hex code	# then 6 digits, RR GG BB
Hexadecimal	Base 16: 0–9 and A–F
RGB range	0 to 255 for each color
White / Black	#FFFFFF / #000000
Equal pairs	A shade of gray
RGB stands for	Red, Green, Blue